Machine Learning

07-04-26

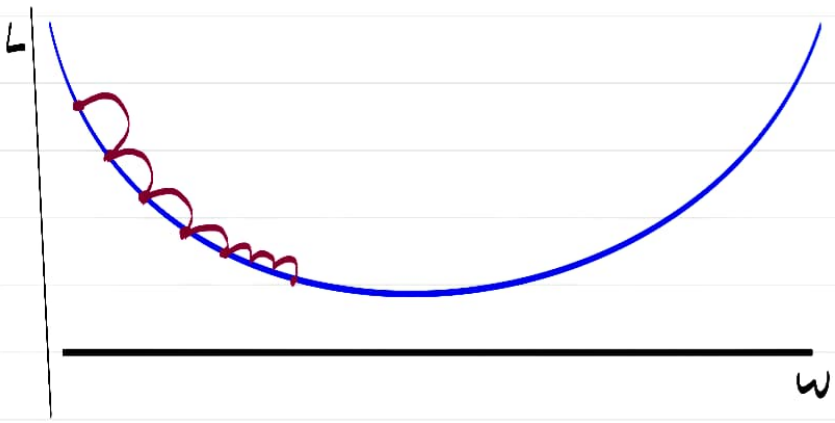
Hyperparameter Tuning

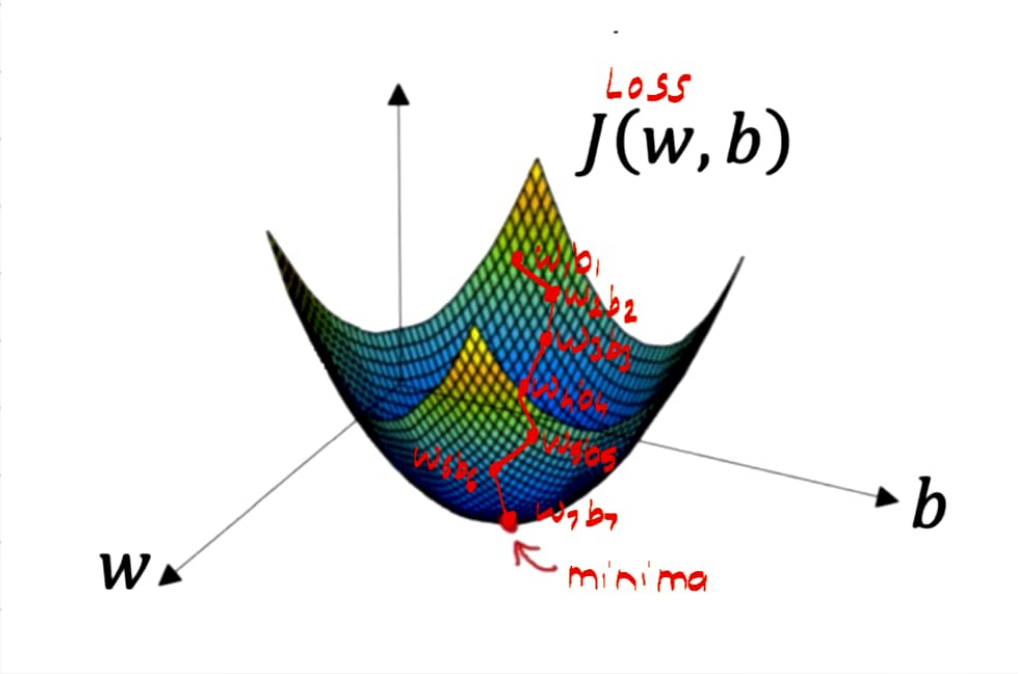
- try and test multiple values of hyperparameter and the value giving best model performance.

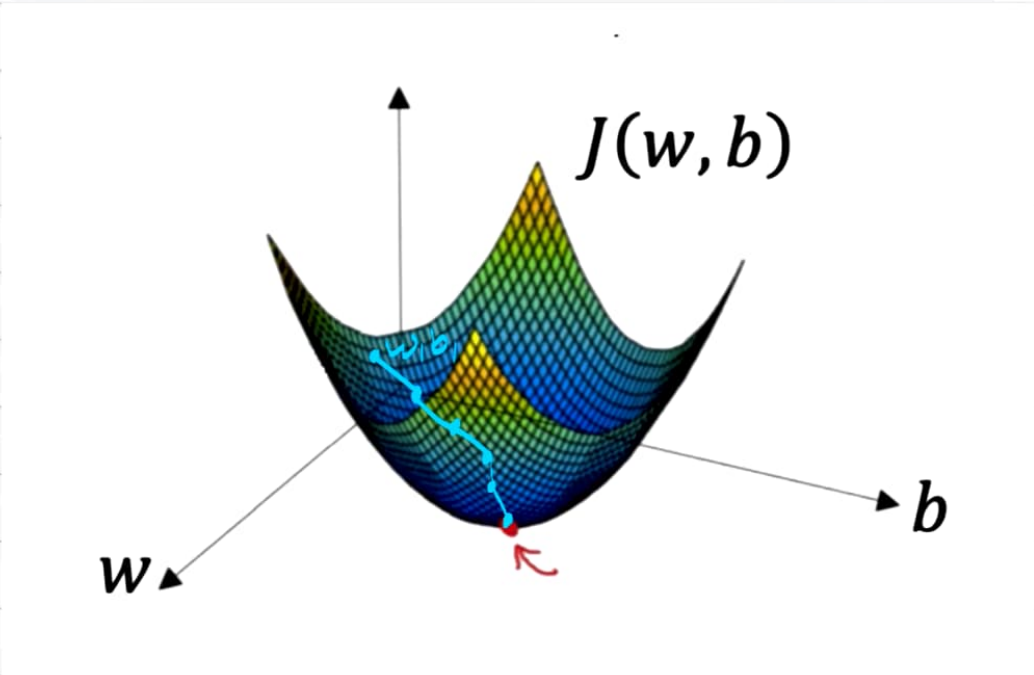
eg. for linear regr. try out multiple values of learning rate $\alpha = 0.1, 0.01, 0.02, 0.04$

Benefits of α learning rate

- makes training process and gradient descent stable
- takes gradual steps towards minima
- avoids overshooting minima



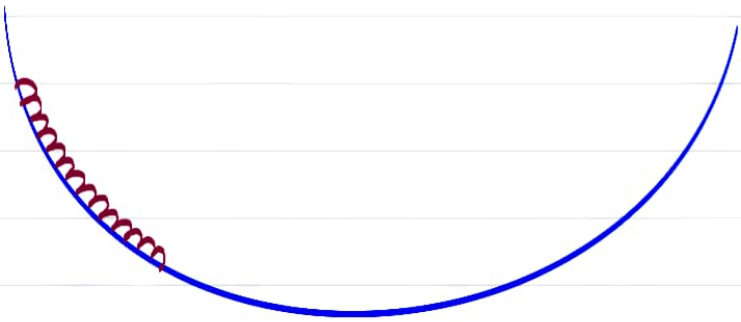


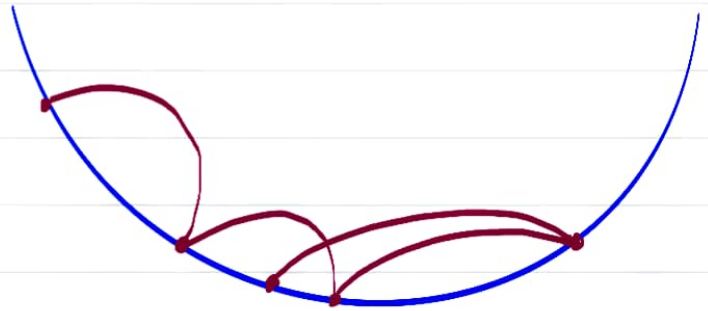


- The process of moving towards minima is called convergence
- model converges when we reach minima.
- We train model until it converges

Learning rate α should be sufficiently small

α - should not be too small or too large



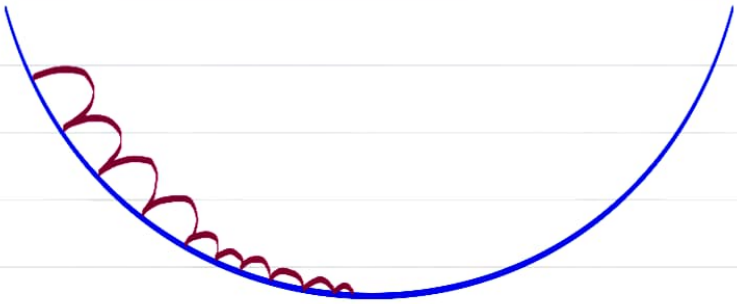


when α is too small

- training becomes really slow
- too much time to reach minima

when α is too large

- unstable training
- overshooting minima



when α is sufficiently small

- stable training

eg. hyp. function : $y = wx + b$

Loss function : $L = f(w, b)$

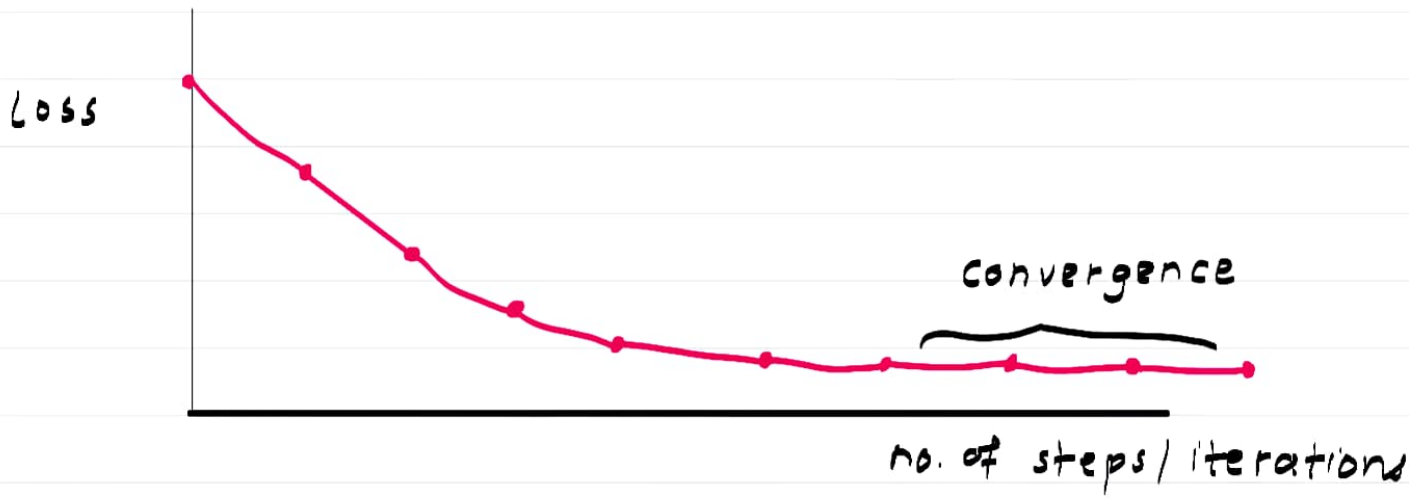
$$L = 5w^2 + 2w + 3wb + b^2 - 2b + 4$$

$$\frac{\partial L}{\partial w} = 10w + 2 + 3b \qquad \frac{\partial L}{\partial b} = 2b - 2 + 3w$$

Start model training

steps	w	b	L	$\partial L/\partial w$	$\partial L/\partial b$
1	w_1	b_1	L_1	$\partial L/\partial w_1$	$\partial L/\partial b_1$
2	$w_2 = w_1 - \alpha \frac{\partial L}{\partial w_1}$	$b_2 = b_1 - \alpha \frac{\partial L}{\partial b_1}$	L_2	$\partial L/\partial w_2$	$\partial L/\partial b_2$
3	$w_3 = w_2 - \alpha \frac{\partial L}{\partial w_2}$	$b_3 = b_2 - \alpha \frac{\partial L}{\partial b_2}$	L_3	$\partial L/\partial w_3$	$\partial L/\partial b_3$
⋮	⋮	⋮	⋮	⋮	⋮

iterations will continue until model converges



Simple Linear Regression

$$y = wx + b$$

x - feature

w, b - model parameters

Multi Linear Regression

$$y = w_1 x_1 + w_2 x_2 + w_3 x_3 + b$$

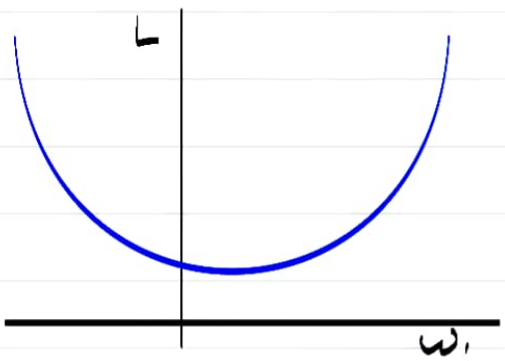
3 features - x_1 x_2 x_3

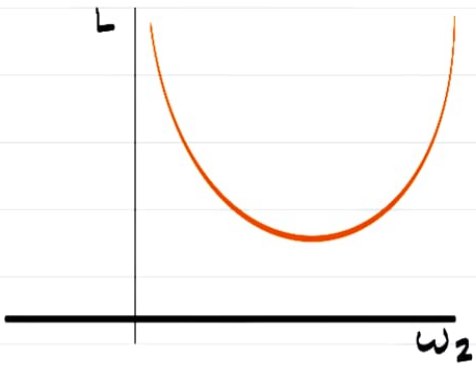
4 model parameters - w_1 w_2 w_3 b

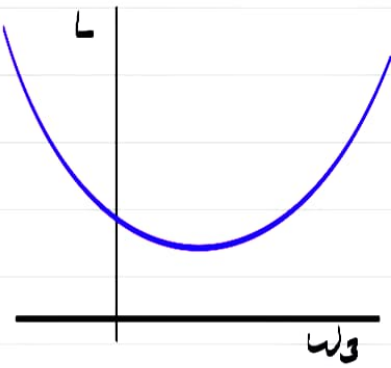
in multilinear regr as well, reln betn loss and model parameters is quadratic and convex

$$L = MSE = \frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

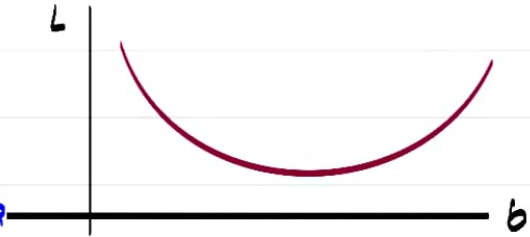
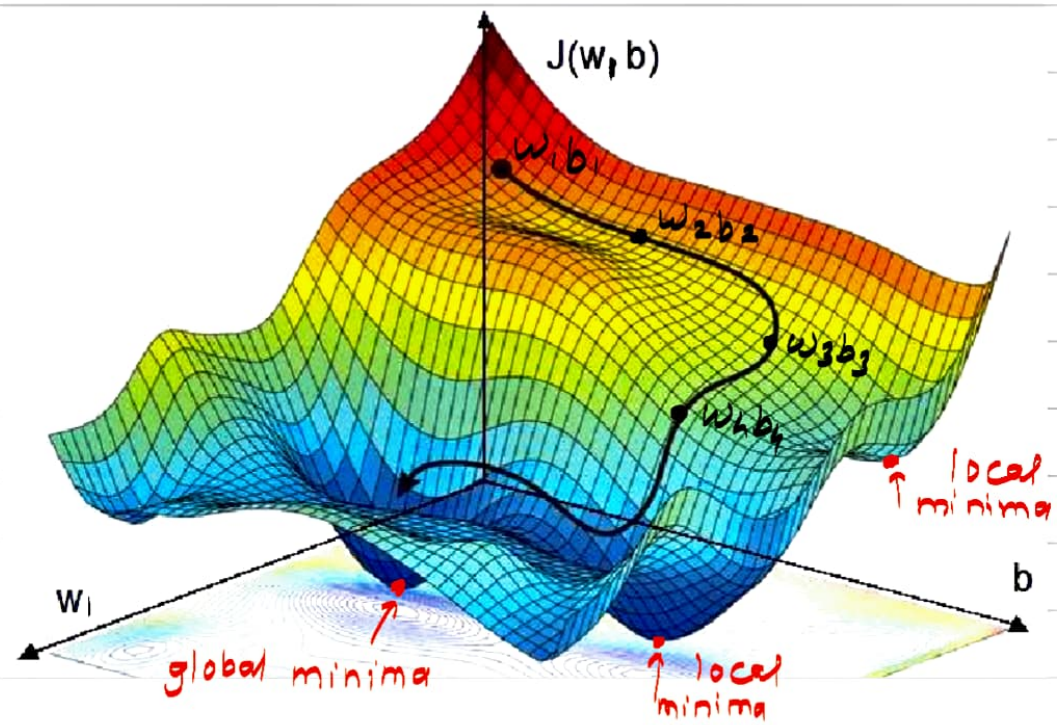
$$= \frac{1}{n} \sum_{i=1}^{n} \left[y_i - (w_1 x_{i1} + w_2 x_{i2} + w_3 x_{i3} + b) \right]^2$$







reln betn L and w/b is convex hence gradient descent will be used to find minima



in multilinear regr. loss function curve looks like this